



The Health-Oriented Transportation Model: Estimating the health benefits of active transportation

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ABSTRACT

Introduction: Overdependence on gasoline-powered personal automobiles in industrialized urban settings has resulted in transportation behaviour that is detrimental to public health. Risk not only stems from an increase in air pollution, but also, and more significantly for wealthy nations, from a reduction in physical activity. Tools and models that demonstrate the magnitude of the health benefits of physical activity are needed to inform policies addressing the epidemic of physical inactivity and to help promote environmentally sustainable cities.

Methods: The Health-Oriented Transportation (HOT) model is a transparent and easily accessible tool that allows users to assess the current and potential health benefits of active transportation (walking or cycling) using data from a one-day travel survey. As a case-study, we apply the HOT model to the population of London, England, using the London Travel Demand Survey.

Results: We estimate that in the 2016 adult population of London, 1,618 and 2,720 deaths were averted in the inner and outer boroughs, respectively, due to transportation-related physical activity; an additional 364 and 946 deaths are potentially averted if the proportion of the adult population that walked or cycled at least weekly from 80% and 73% to 100% in the inner and outer boroughs, respectively. A 50% increase in walking/cycling mode share among active travellers would result in a 2.5% reduction in premature deaths in the adult population of London.

Conclusions: Estimating the potentially large health benefits arising from active transportation in an urban setting can provide key public health information for urban planners and local officials, informing investments in infrastructure and critical public health programs.

1. Introduction

The ubiquity of gasoline-powered personal automobiles has led to urban transportation behaviour that is both detrimental to public health and environmentally unsustainable. This phenomenon is directly related to increasing air pollution and, importantly especially for industrialized countries, to a reduction in physical activity (Woodcock et al., 1943; Anenberg et al., 2019). Estimates are that by 2050, 70% of the world's population will live in urban environments (Birch and Wachter, 2011). In addition to improving the sustainability of urban environments, alternative urban transportation can promote exercise behaviour. Quantifying the extent to which physical activity may be increased is essential to informing policy and public investment in the built urban environment that can lead to healthier populations and cities. Physical inactivity is a well-known modifiable risk factor for many chronic diseases including diabetes, colon cancer, obesity, hypertension and depression (Warburton et al., 2006). Inadequate levels of physical activity increase the risk of these diseases by 20–30%, thereby decreasing quality of life, and shorten lifespan by 3–5 years (WHO). A 2012 worldwide analysis of the effects of physical activity on major non-communicable diseases conducted by Lee et al. estimated that inactivity causes 9% of premature mortality globally (Lee et al., 2012). Nonetheless, the world's population is experiencing a decline in physical activity, leading to what some have referred to as an *epidemic of inactivity* (Bauman et al., 2012). Lear et al. estimate that 23% of the world's population is insufficiently active (Lear et al., 2017). To combat this growing global health crisis, in 2013 the World Health

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Organization recommended decreasing the rate of insufficient physical activity by 10% by 2020 (World Health Assembly, 2013). As such, public health officials must find ways to foster physical activity at the population level.

We aim to quantify the potential health benefits arising from travel scenarios that facilitate more walking and cycling among a population as part of their routine transportation. First, we model the health benefits associated with current transportation behaviour as reflected in a one-day transportation survey. The Health-Oriented Transportation (HOT) model is a statistical model that estimates the change in health outcomes associated with a change in levels of physical activity due to transportation. Any transportation requiring significant physical activity is considered active transportation, although the HOT model currently focuses exclusively on walking and cycling. Using detailed travel survey data from London, England we examine the relationship between physical activity and all-cause mortality as a case study for estimating current and potential health benefits of active transportation. The HOT model is accessible via a web-based application to which the user can upload any suitably formatted travel survey data for analysis. This work is part of the Complex Urban Systems for Sustainability and Health (CUSSH) project, created to assist cities in making transformative change towards global health goals (Belesova et al., 2018).

2. Materials & methods

Although recreational exercise is a common way to achieve physical fitness in high income countries, the significant time and/or financial commitments required are a barrier for many. More accessible and routine forms of physical activity, such as commuting via active transportation, are thus a priority for population-wide health (Warburton et al., 2006). For this reason, walking and cycling as regular means of transportation are the focus of the HOT model. The physical activity achieved from walking or cycling for transportation is referred to as *travel activity*. The HOT model estimates the actual distribution of travel activity in the population and hypothetical distributions that represent new travel scenarios in order to then estimate the change in the health outcome of interest. In this analysis, we estimate the proportional change in all-cause mortality for two travel scenarios representing the current and potential health effects of active transportation. The HOT model is modular by design and the outcome of interest may be changed, provided an exposure-response function is available for the given outcome. For more discussion on the exposure-response function see the online supplement.

2.1. Travel activity

We describe the statistical distribution of travel activity in a population in terms of three parameters: participation, frequency, and intensity, representing the number of people who walk or cycle, how often, and the amount of physical activity achieved in the process, respectively. Typically, one-day travel survey data are used to study travel behaviour in a population and we therefore tailor our model to such surveys. However, studies that estimate the health benefits of physical activity usually define exposure using a weekly rate of physical activity. To accommodate these differing time scales, we must extrapolate the one-day estimate to a weekly rate.

We begin with an unadjusted daily travel activity estimate computed as the weighted sum of walking and cycling duration in hours, as recorded in a one-day travel survey.

$$\tilde{T} = \alpha_w T_w + \alpha_c T_c$$

The parameters α_w and α_c represent the strenuousness of walking and cycling, respectively. In the HOT model we use $\alpha_w = 3$ MET and $\alpha_c = 6$ MET, where MET (metabolic equivalent task) is the unit used in Ainsworth et al. (Ainsworth et al., 2011). T_w and T_c represent the total one-day duration of walking and cycling in hours. For reasons described above, the intensity of travel is measured as a weekly rate, and therefore $\tilde{T}$ must be scaled by the expected number of active days per week, $7f_1$, where f_1 is the mean daily frequency of active transportation, $T = 7f_1 \tilde{T}$. The distribution of travel activity in the population is modelled as a mixture of zero (inactive

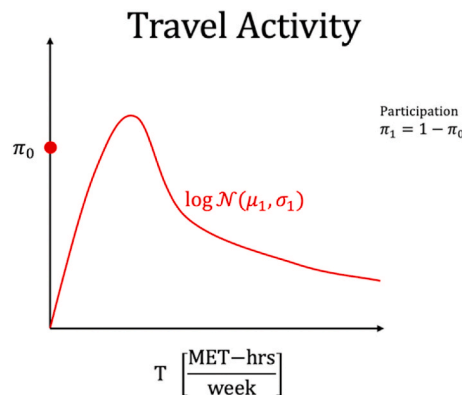


Fig. 1. The distribution of travel activity in the HOT model. The distribution is a mixture with probability π_0 that travel activity is zero and probability π_1 from a log-normal distribution. π_1 is known as the participation rate and μ_1 the intensity.

travellers) and log-normal with mean and standard deviation (μ_1, σ_1) estimated from the data (active travellers). We provide a representation of the model in Fig. 1 in which we see that a proportion of the population, π_0 , obtains zero travel activity (T) represented by the large dot on the vertical axis. The travel activity obtained by the remainder of the population is modelled as log-normal with parameters μ_1 and σ_1 and we refer to μ_1 as the *intensity* of travel activity. The *participation* in active transportation is represented with $\pi_1 = 1 - \pi_0$.

Note that intensity, μ_1 , is a conditional mean; only subjects with non-zero travel activity are included. Individuals with zero travel activity are accounted for in the participation parameter. Participation is defined as the proportion of the population who engage in active transportation, and thus a threshold is required to define the minimum frequency of active transportation needed to be considered as engaging in active transportation. In this analysis we use an estimate of the proportion who self-report as walking or cycling *at least weekly*. At least one trip of a minimum of 10 min of walking or any amount of cycling is required to be defined as having walked or cycled, respectively (short walks to the car are not counted). On any given day there will be some individuals who participate in active transportation at least weekly who do not walk or cycle on that day and so the proportion of individuals with some travel activity on the survey day must be less than the value for participation. We define the frequency of active transportation, f_1 , as the mean daily frequency at which active travellers are active. For a given individual a frequency of one means the traveller walks or cycles every day, and so the true population-wide estimate of frequency must be less than one, for certainly not everyone is active every day. And since this frequency is less than one, the prevalence, or proportion of the population who engage in active transportation on a given day, is a biased estimator of participation. The correction factor is the inverse of f_1 , and so $f_1 = p_1/\pi_1$, where p_1 is an estimate for prevalence. (See Appendix for details.)

2.2. Current and potential health benefits

For the estimation of current and potential health benefits of active transportation we conduct two comparative risk assessments. We consider the cases of zero and full participation in active transportation. By increasing the amount of travel activity in the baseline model, exposure to more physical activity is increased, and, in the HOT model, exposure to physical activity decreases the risk of the response, namely premature death. The HOT model uses an exposure-response function that estimates the relative risk of all-cause mortality, given a level of physical activity. We focus on all-cause mortality in this analysis for simplicity, however, exposure-response functions for disease risk can be used as well. Estimates of the change in disease incidence would allow the disease burden to be characterized in terms of measures such as YLD, YDD and DALY. See the online supplement for more discussion on the choice of exposure-response function. The relationship between physical activity and all-cause mortality is modelled using the hazard ratio functions of (Arem et al., 2015). See the online supplement for more details on the exposure-response function. With an exposure-response function, along with the distributions of travel activity at baseline and at a given scenario, we can estimate the population attributable fraction, ρ , defined as the proportional change in expected risk between a baseline and an alternative scenario, as follows (Rockhill et al., 1998).

$$\rho = \frac{\int R(x)Q(x)dx - \int R(x)P(x)dx}{\int R(x)P(x)dx} \approx \frac{\sum_j R(x'_j) - \sum_j R(x_j)}{\sum_j R(x_j)}$$

In the above formula: P and Q represent the distribution functions for physical activity at baseline and scenario, respectively; R is a function representing the risk of all-cause mortality in terms of physical activity; and x_j and x'_j represent equally spaced quantiles of the baseline and scenario distributions of physical activity, respectively. The participants in the survey used by Arem et al. were surveyed only about recreational activities, and so for the HOT model we assume that recreational physical activity remains constant across transportation scenarios. We compute the exposure level in the scenario distribution by adding (or subtracting) the additional physical activity (PA) gained (or lost) through commuter travel activity, δ_{TA} .

$$PA_{\text{scenario}} = PA_{\text{baseline}} + \delta_{TA}$$

The physical activity at baseline, PA_{baseline} , is simulated using a distribution derived from the results of Arem et al. (Arem et al., 2015). See Appendix for more details on the baseline distribution of physical activity. We choose to use physical activity as the exposure variable in the HOT model primarily because it makes for a more parsimonious model. Two activities, walking and cycling, are reduced to one value: the physical activity due to travel, or *travel activity*. HOT then models the *change* in travel activity due to alternative walking and cycling behaviour. This *change* in physical activity may be added to any type of baseline physical activity, e.g., leisure-time or total, if we assume that walking and cycling has the same physiological effect on health as the baseline physical activity type. In the analysis presented here we use leisure-time physical activity and the distribution of Arem et al. (Arem et al., 2015), however, total physical activity could be used as well, as in Lear et al. (Lear et al., 2017). The HOT model considers gains in travel activity to have the same physiological effect on health as gains in recreational physical activity. If we consider a hypothetical scenario in which all active travellers cease being active, keeping other physical activity behaviour constant, and we estimate the corresponding increase in all-cause mortality using the HOT model, we then have an estimate for the current health benefits of active transportation ($\pi_0 \rightarrow 1$) expressed in terms of premature deaths averted. It is difficult to place an upper limit on the amount of travel activity achievable by a population, and so for potential health benefits we use a scenario in which the entire population of interest participates at the current (baseline) levels of travel activity intensity, i.e., $\pi_0 \rightarrow 0$ with μ_1 and σ_1 constant.

2.3. Data description

The primary data source for the case study of London, England is the London Travel Demand Survey (LTDS). The LTDS is a continuous household-based travel survey of the London area conducted by Transport for London starting in 2005, (Transport for London, 2019). We present results for 2016. The annual sample is approximately 8,000 households with the response rate for full interviews ranging from 49% to 54% (Fairnie et al., 2016). The LTDS is made up of three questionnaires led by a qualified interviewer, in-person, at the respondent's home: a household questionnaire which collects basic demographic data, an individual questionnaire completed by all members of the household at least five years of age which collects further demographic data as well as travel-related information, and, lastly, a trip sheet, completed by every member of the household at least five years of age that gathers data on all trips made on a specified day not necessarily in the workweek (Fairnie et al., 2016). For a more detailed description of the LTDS data see (Fairnie et al., 2016). The total walk and cycle duration were computed by summing reported duration over all walk and cycle stages, where a stage is defined as a component of a trip, e.g., a three-stage trip could be a walk to a transit stop, followed by the transit ride, followed by a walk from the transit stop to the destination. Trips with purpose recorded as "Leisure trip – enjoyment" were excluded since our focus is on transportation behaviour and these trips are, presumably, purely recreational. For analyses stratified by age, cut-offs of 18 and 65 years were used to define children and seniors. Borough-specific estimates for participation were taken from the results of a survey performed in 2016-17 by the UK Department for Transport in which participants reported the approximate frequency of walking or cycling (UK Department for Transport, 2019). This additional data source is required to estimate the frequency of active transportation. If estimates for participation were not available, we would assume the frequency is constant and estimate the participation. We used the inner/outer classification of boroughs used by the London Council. Participation estimates for this analysis use the proportion of people who self-report to walk or cycle at least weekly. We removed subjects with less than 1 or greater than 50 MET-hrs./week of travel activity. A subject estimated to have less than 1 MET-hrs./week of travel activity reported the equivalent of approximately 4 min of walking on the travel survey day. With reporting bias this is likely to correspond to a few short walks to the car, and not considered sufficient for our purposes. An effort of over 50 MET-hrs./week corresponds to cycling for approximately an hour and a half on the day of the survey.

3. Results

In the Greater London Metropolitan Area, active transportation participation is estimated to be around 0.75. Fig. 2 displays the values for participation across London, which range from 0.64 in Barking and Dagenham to 0.89 in the City of London. In general, boroughs in the inner part of London tend to participate more in active transportation, with the inner boroughs' participation at around 80% and the outer boroughs 73%. (See online supplement for demarcation of inner/outer boroughs.)

We see in Fig. 2 that not only does the participation decrease in the outer boroughs, but so too does the frequency. In the outer boroughs, the frequency of active travellers is, on average, 68%, while in the inner boroughs it is 79%. The mean frequency of active transportation varies across London boroughs from approximately four to six days per week. A strength of the HOT model is that it adjusts intensity estimates based on this frequency. Failing to capture this variability could lead to misestimation of the weekly rate of physical activity.

To estimate the current and potential health benefits of active transportation we focus on the adult population of London in 2016, stratified by inner/outer borough status. In 2016 there were an estimated 14,470 and 34,454 deaths among the adult population (18–65 years old) in the inner and outer boroughs of London, respectively (Death Registrations, England & Wales, 2020). If the health benefits of active transportation were eliminated altogether through a complete lack of participation in walking and cycling there would be an 11.2% and 7.9% increase in the mortality rate among the inner and outer boroughs, respectively. Correspondingly, an estimated 1,618 and 2,720 deaths are averted annually among residents of London's inner and outer boroughs through current levels of active transportation. The remaining potential (and achievable) health benefits of active transportation may be estimated by increasing the model participation proportion to one, while maintaining current levels of intensity. If this were to happen, the current

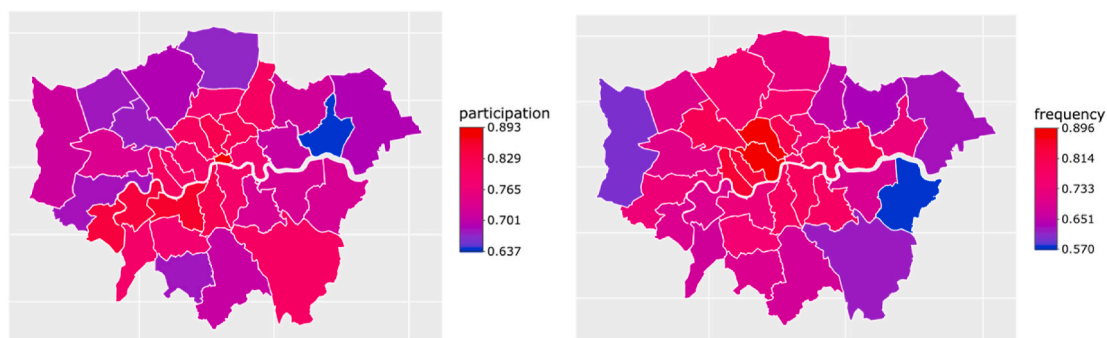


Fig. 2. Participation of active transportation (left) and daily frequency of active transportation (right) in London, England, stratified by borough. Participation estimates are from 2016-17. Note that the estimates for frequency use prevalence estimates for all twelve years, 2005–2016.

mortality rate would decrease by 2.5% and 2.9% in the inner and outer boroughs respectively, corresponding to 364 and 946 deaths averted annually. Table 1 displays the reduction in all-cause mortality for the two scenarios described above along with six scenarios defined by an increase in walking and cycling. It can be seen through inspection of Table 1 that increasing participation is more beneficial than increasing intensity. Maintaining and increasing levels of participation in active transportation should therefore be a priority. This includes working towards eliminating obstacles that discourage current active travellers, as well as continuing to try to find ways to facilitate new active travellers. We see that in order to achieve health benefits equivalent to full participation at current levels of intensity, the intensity of current active travellers in London must be increased by 50%. In Fig. 3, the estimated reduction in premature death is shown in terms of the percent increase in intensity. An increase in intensity is equivalent to an increase in both walk and cycle mode share. Travel scenarios expressed in terms of mode share can be characterized in terms of an increase in intensity. In Fig. 3, we assume no change in participation. For scenarios in which the participation is varied, see the Additional Figures section of the online supplement. No significant change can be seen for increases in intensity less than 5% due to the variance of the distribution of travel activity.

4. Discussion

The global epidemic of inactivity, largely brought about by dependence on motorized vehicles for daily, routine transportation, represents a significant threat to public health, and is expanding across most regions of the world. Many studies have investigated the positive impact on air quality and health outcomes related to air pollution resulting from a decrease in motorized transportation, but few have analysed the health gains brought about by a concomitant increase in physical activity. Although the current health benefits of active transportation are far more substantial than many expect, this information is not often leveraged in the creation of transportation policies. The health impact of transportation policies aimed at decreasing dependence on motorized vehicles as a means to improve air quality and mitigate climate change would be significantly expanded by taking into account the extent to which fostering active transportation can combat the epidemic of physical inactivity and improve public health. Our goal is to create a model that demonstrates the magnitude of the population-level health benefits achieved from lowering *built-in* dependence on personal, motorized transportation.

Many health impacts assessments of active transportation have been performed and difficulty arises when comparing across the studies due to varying model assumptions (Mueller et al., 2015). The creation of models and tools that are well-defined and available to the public is an important step towards combatting this difficulty. Multiple implementations of the Integrated Transportation and Health Impacts Model (ITHIM) have been published using various platforms including R, Analytica and Microsoft Excel (Woodcock et al., 2013; Wu et al., 2019). See the online supplement for more discussion of ITHIM implementations. We are aware of three tools for health impacts assessment of active transportation available publicly; The Health Economic Assessment Tool for cycling and walking (HEAT), the Impacts of Cycling Tool, and the Integrated Transportation and Health Impacts Model (ITHIM) Tool (HEAT, 2017; Woodcock et al., 2018; Maizlish et al., 2017). The HEAT tool and ITHIM implementations both incorporate health outcomes due to air pollution and road injuries. The HOT model currently only estimates health outcomes due to changes in physical activity. The R package, discussed below, is modular by design so that analysis can be extended to include other health components. We are currently working to develop additional modules. Until then, the HOT model serves as an open-source alternative to the physical activity modules of HEAT and ITHIM. HEAT, maintained by the World Health Organization, is a web application focused on the economic effects of cycling and walking. The Impacts of Cycling Tool (ICT) explores scenarios of cycling uptake, similar to HOT's participation parameter, providing estimates of resulting emissions, travel patterns, and health impacts (Woodcock et al., 2018). The HOT method includes an individual's participation in walking or cycling, representing active travel, while the ICT is solely concerned with regular (weekly) engagement in cycling. Additionally, the ICT examines the trip replacement involved in cycling scenarios, considering e-bike availability and equitable distribution of new cycling trips. Like HOT and HEAT, the ICT is accessible as a web application and may be used to effectively communicate to policymakers, demonstrating the benefits of a modal shift to active modes of travel. The ITHIM Tool is an application that may be run locally using RStudio and the *shiny* R package, and there is an additional online ITHIM USA web application for use applying ITHIM methods to the United States. HOT is the first R package and one of three web applications for health impacts assessment of active transportation. Unlike the other applications mentioned above, the HOT R package and web application work directly with travel survey data and can be easily incorporated into more general analyses of travel survey data.

4.1. R package and web application

The goal of this work is to provide the academic community, as well as the public, with a well-defined, accessible way of evaluating

Table 1

The reduction in all-cause mortality for eight scenarios; zero/one hundred percent participation in active transportation, and five scenarios defined by an increase in walking and cycling. For the six scenarios based on increased walking and cycling, the participation was held constant at the baseline value of 0.75.

	Participation		Increase in walking and cycling					
	0%	100%	5%	10%	25%	50%	100%	200%
Inner London	–11%	3%	<1%	1%	2%	4%	7%	10%
Outer London	–8%	3%	<1%	1%	2%	3%	5%	8%

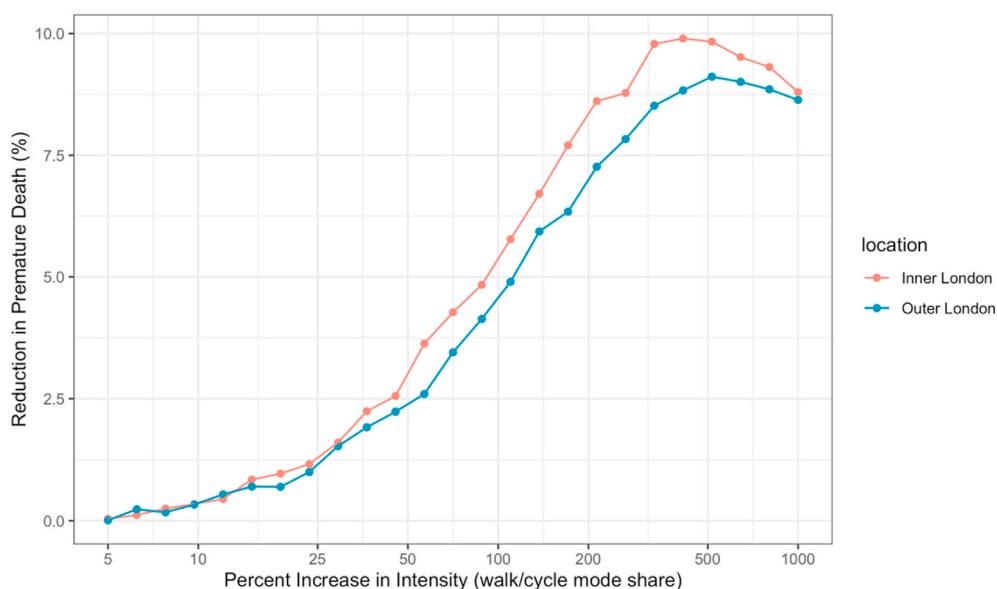


Fig. 3. The reduction in premature death in terms of the percentage increase in intensity of active transportation. Increasing the intensity is equivalent to increasing both the walk and cycle mode share among active travellers. Note that the value for participation is kept at the baseline value in the alternative scenario.

the health impacts of walking and cycling so that awareness of the extent of the problem is increased. To achieve this we required a standardised, accessible and extensible way of evaluating the health impacts (HOT model) and a simple tool that requires only travel survey data and no advanced analytical skills. In order that the HOT model be accessible and extensible, we have created an R package and made it publicly available on *GitLab*. See online supplement for all URLs. This package contains the functions used in this analysis along with documentation. In this way, the HOT method can be examined by other researchers and extended if desired. The package has been designed to be flexible by using a modular design in which components of the model can be replaced, e.g., the exposure-response function. A web application has been developed as a simple tool requiring travel survey data and optional participation data. The HOT web application processes travel survey data directly, unlike HEAT which requires the user to first estimate parameters externally. To use the web application the user needs only a properly formatted travel survey data set and, optionally, estimates for participation. The web application allows users to run functions such as *getIntensity*, *getPrevalence*, and *getCRA* used to parameterize and then analyse the HOT model in a point-and-click environment. Currently the web application only displays results for two scenarios, zero and full participation, but will soon have the ability to analyse other scenarios.

It is our hope that experienced users will be able to quickly learn and apply the package through internal package documentation and tutorials, and ultimately incorporate it into their own analyses and expand on the methodology. For users that would prefer a simpler interface, we have created a web application that allows users to run basic functions of the package. The web application tool is under continued development and is available to be customized to better suit user needs.

4.2. Limitations

As with all models, the HOT model is subject to the limitations inherent to the input data, in this case one-day travel surveys, which suffer from recall bias, over/under-reporting of activity, financial expense, and many others. The HOT model can estimate the correct inflation factor if estimates for participation are known. It is often the case that participation estimates are not available. When participation estimates are unknown, the HOT model uses a constant value for frequency. For this case study participation estimates were available, however the estimates included individuals who walk or cycle for recreational reasons. Currently in the HOT model the participation parameter is held constant across age and sex. Similarly, the distribution of physical activity is not age or sex dependent and the default used by the HOT model may not accurately reflect the true distribution in the population. Over-estimates of the health effects might occur in populations with greater than average physical activity levels. Future versions of the HOT model will allow for greater control over both the distribution of physical activity and participation in active transportation. The HOT R package does not currently provide built-in methods for uncertainty analysis. Three parameters are required to specify the HOT model at the baseline and alternative scenarios (participation, intensity and variance of travel activity). Baseline estimates of these parameters are computed using survey data and will therefore be subject to variability. We intend to incorporate the ability to specify confidence intervals for each of the parameters so that the model uncertainty can be evaluated as part of the R package. In the meantime, users can perform the analysis manually, using the endpoints of each of the confidence intervals to determine the degree of uncertainty.

4.3. The epidemiology of physical activity

It is important to note that the HOT model is concerned with modelling changes in physical activity and therefore may be viewed as a physical activity model that works with travel data, comparable to the physical activity modules of other more comprehensive health impact models. Physical activity modelling has shown that age, sex, health status, self-efficacy, motivation and even genetic makeup are all associated with physical activity (Bauman et al., 2012). In this manuscript we investigate *What If* travel scenarios and not in how these scenarios may be achieved. As such, we need not account for how subpopulations have varying levels of physical activity. The HOT model, therefore, uses an unconditional distribution of physical activity derived from the physical activity modelling of Arem et al. (Arem et al., 2015). The health impacts of including additional walking and cycling trips in an individual's physical activity regimen is highly heterogeneous and clearly depends on what activity is displaced. An isotemporal substitution paradigm was developed as a new analytic model to study the time-substitution effects of one activity for another, and although the substitution paradigm is beyond the scope of the HOT model, it is important to remember that the alternative travel scenarios presented here are hypothetical scenarios in which no physical activity is displaced by new walking or cycling trips (Mekary et al., 2009). As with the travel survey data presented here, physical activity models often rely on physical activity questionnaires. Self-reported data is subject to response bias, and in the case of physical activity, the difficulty of standardising measures across studies. It is generally thought that individuals tend to overestimate participation in vigorous activities and underestimate participation in light to medium intensity activities (Sallis and Saelens, 2000).

5. Conclusions

It is important for public health officials, city officials and active transportation advocates to have access to estimates of the positive health effects of walking and cycling and for this information to be incorporated in advocacy related to vehicle emissions and climate change. There is a clear need for tools and models to generate high-quality estimates, without being overly dependent on data that are often difficult to obtain. The HOT model, R package and web application provide a well-defined method along with a publicly available toolkit that enables people from across all sectors to perform a rapid assessment of the public health benefits of active transportation given merely the results of a one-day travel survey.

CRediT authorship contribution statement

Samuel G. Younkin: Conceptualization, Data curation, Formal analysis, Methodology, Software, Project administration, Writing – original draft. **Henry C. Fremont:** Visualization, Software, Formal analysis, Writing – review & editing. **Jonathan A. Patz:** Supervision, Writing – review & editing.

Declaration of competing interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jth.2021.101103>.

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